

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

DR. SQUATCH, LLC,
Petitioner,

v.

THE PROCTER & GAMBLE COMPANY,
Patent Owner.

IPR2024-01173
Patent 10,905,647 B2

Before SHERIDAN K. SNEDDEN, SUSAN L. C. MITCHELL, and
CHRISTOPHER M. KAISER, *Administrative Patent Judges*.

MITCHELL, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. *Background and Summary*

Dr. Squatch, LLC (“Petitioner”) filed a Petition requesting *inter partes* review of claims 1–7 and 9–19 of U.S. Patent No. 10,905,647 B2 (Ex. 1001, “the ’647 patent”). Petition (“Pet.”), Paper 1. The Procter & Gamble Company (“Patent Owner”) filed a Preliminary Response identifying itself as the owner of the ’647 patent. Preliminary Response (“Prelim. Resp.”), Paper 7.

We instituted trial on January 29, 2025. Paper 12 (“Inst. Dec.”). During trial, Patent Owner filed a Patent Owner Response. Paper 16 (“PO Resp.”). Petitioner filed a Reply (Paper 20 (“Pet. Reply”)), and Patent Owner filed a Sur-reply (Paper 25 (“PO Sur-Reply”)). We held an oral hearing on October 22, 2025, and a transcript of that hearing is of record. Paper 34 (“Tr.”).

We have jurisdiction under 35 U.S.C. § 6(b). After considering the full record developed through trial, we determine that Petitioner has proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

B. *Real Parties-in-Interest*

Petitioner identifies Dr. Squatch, LLC, as its real party-in-interest. Pet. 1. Patent Owner identifies The Procter & Gamble Company as its real party-in-interest. Paper 5, 1.

C. *Related Matters*

The parties indicate that the ’647 patent is involved in *The Procter & Gamble Company v. Dr. Squatch LLC*, Case No. 2-24-cv-04711 (C.D. Cal.),

filed November 29, 2022. Pet. 1; Paper 5, 1; Paper 33, 2. Petitioner has also filed petitions for *inter partes* review of the following patents: IPR2024-01104 challenging U.S. Patent No. 11,540,999; IPR2024-01105 challenging U.S. Patent No. 10,966,915; and IPR2024-01174 challenging U.S. Patent No. 11,497,706. Paper 4, 1; Paper 33, 2.

D. The '647 Patent (Ex. 1001)

The '647 patent is titled “Antiperspirant and Deodorant Compositions.” Ex. 1001, code (54). The '647 patent is directed to antiperspirant and deodorant compositions and methods of making the compositions. *Id.* at 1:15–16, 10:65. More specifically, the '647 patent is directed to formulating, in response to consumer demands, an aluminum-free, natural fragrance deodorant that provides low irritation, while maintaining sufficient odor protection. *Id.* at 1:41–43.

The '647 patent explains that

[c]onsumers are also seeking more natural products complete with fragrances that are mostly natural or essential oil based. They are further seeking products with lower irritation than they have experienced with baking soda based products. The challenge with formulating with the natural and essential oil fragrances is that they can be less stable in the presence of heat and extreme pH (either high or low). And products formulated with baking soda, which has a relatively high pH and high water solubility, can increase irritation, even for consumers with moderate sweat rates. Highly water soluble alkaline powders contribute negatively towards natural or essential oil stability as well, especially in hot process[es] needed to melt waxes. And high water soluble powders can also lead to gritty product, as the water soluble powders can agglomerate when exposed to moisture released from powders during the hot batch process.

Id. at 1:25–40.

The '647 patent further states that the

inventors of the present invention have found, surprisingly, that the combination of piroctone olamine and other antimicrobials can provide significant antimicrobial activity against two of the most common underarm odor bacteria *C. mucofaciens* and *S. epidermidis*, which results in consumer odor protection on par or greater than some of the commonly used commercial deodorants available today.

Ex. 1001, 1:48–55; *see id.* at 12:24–34 (describing regulatory problems with using a sufficient amount of only one antimicrobial). To maintain the efficacy of piroctone olamine in the presence of iron and other impurities, the '647 patent states that combining chelators and piroctone olamine in an anhydrous formulation or in particular ratios “can provide significantly higher levels of anti-fungal activity than either material alone.” *Id.* at 1:56–63, 7:48–55 (stating a chelator can reduce the amount of piroctone olamine bonding with the iron impurity).

The '647 patent also observes that antimicrobials may be able to reduce microbes on the skin, but not affect microbes within hair follicles that may contribute to malodor. Ex. 1001, 1:64–67. To solve this problem, “[t]he inventors of the present invention have found that piroctone olamine and other antimicrobial powders, if used at a size that can fit into a hair follicle, can deliver surprisingly superior antimicrobial activity.” *Id.* at 1:67–2:3. “It is believed that antimicrobial powders may provide a better deposition and have more longevity on the skin than antimicrobials delivered in a different form.” *Id.* at 10:61–63; *see id.* at 11:1–21.

According to the '647 patent, “a deodorant stick having at least about 25% of a liquid triglyceride and structurants that have a melting point of greater than about 50° C. can result in a deodorant stick with a hardness from about 70 mm*10 to about 140 mm*10.” *Id.* at 15:7–10. The '647 patent asserts that “[s]uch a deodorant stick is able to comprise consumer-

perceived natural ingredients, while offering a pleasant consumer experience in terms of its hardness.” *Id.* at 15:10–13.

The ’647 patent describes that the antiperspirant/deodorant compositions “can contain a primary structurant, an antiperspirant active, a perfume, and additional chassis ingredient(s) . . . in the form of a solid stick.” *Id.* at 14:47–54. The ’647 patent defines a “solid structurant” as “any material known or otherwise effective in providing suspending, gelling, viscosifying, solidifying, and/or thickening properties to the composition or which otherwise provide structure to the final product form.” *Id.* at 15:58–62.

The ’647 patent further describes a deodorant stick as comprising at least 25% of a liquid triglyceride; a primary antimicrobial having a water solubility of at most about 90 g/L at 25° C; a fragrance composition comprising at least about 50% natural oils, essential oils, or a combination thereof; and a primary structurant with a melting point of at least 50° C.; said deodorant stick being free of an aluminum salt; and said deodorant stick having a hardness from about 70 mm*10 to about 140 mm*10, as measured by penetration with ASTM D-1321 needle.
Id. at 2:11–20.

E. The Challenged Claims

Petitioner challenges claims 1–7, 9–13, and 16–19.¹ Pet. 1. Claims 1, 16, and 18 are independent. Claim 1 includes a hardness limitation that is contested in this *inter partes* review, but claims 16 and 17 that are subject to the anticipation ground involving Lesniak do not. Therefore, we find that claims 1, 16, and 17 are representative and are reproduced below.

¹ Claims 14 and 15 have been statutorily disclaimed. *See* PO Resp. 9 n.1; Ex. 2021, 1. We have adjusted the challenged claims to omit these claims.

1. A deodorant stick comprising:
 - a. at least 25% by weight liquid triglyceride;
 - b. a primary antimicrobial having a water solubility of at most about 90g/L at 25° C.;
 - c. a fragrance composition comprising at least about 50% by weight of natural oils, essential oils, or a combination thereof; and
 - d. a primary structurant with a melting point of at least about 50° C.; and

said deodorant stick being free of an aluminum salt; and

said deodorant stick having a hardness from about 70 mm*10 to about 140 mm*10, as measured by penetration with ASTM D-1321 needle.

Ex. 1001, 27:20–33.

16. A deodorant stick comprising:
 - a. at least 60% by weight liquid triglyceride;
 - b. at least one antimicrobial;
 - c. a fragrance composition comprising at least about 50% by weight of natural oils, essential oils, or a combination thereof; and

said deodorant stick being free of an aluminum salt.

17. The deodorant stick of claim 16, further comprising a primary structurant having a melting point from about 50° C. to about 80° C.

F. Evidence

Petitioner relies upon information that includes the following.

Ex. 1003, Lesniak et al., US 11,433,018 B2, issued September 6, 2022 (“Lesniak”).

Ex. 1004, Phinney et al., US 9,314,412 B2, issued April 19, 2016 (“Phinney”).

Ex. 1006, Bianchi et al., US 6,048,518 B2, issued April 11, 2000 (“Bianchi ’518”).

Ex. 1007, Wayback Machine archives, captured May 26, 2016, November 23, 2016, and November 1, 2015 respectively for: <https://www.nativecos.com>, <https://www.nativecos.com/product/deodorant/>, and <https://www.nativecos.com> (“Native”).

Ex. 1008, Bianchi, US 2007/0166254 A1, published July 19, 2007 (“Bianchi ’254”).

Ex. 1009, Wayback Machine archive, captured January 26, 2017, for <http://soapdelinews.com/2016/11/easy-homemade-deodorantrecipe.html> (“Easy Homemade”).

Ex. 1067, Nabavi et al., *Antibacterial Effects of Cinnamon: From Farm to Food, Cosmetic and Pharmaceutical Industries*, 7 NUTRIENTS 7729–48 (2015) (“Nabavi”).

Petitioner also relies on the declarations of Bozena B. Michniak-Kohn, Ph.D. (Exs. 1002, 1086) to support its contentions.

Patent Owner relies on the declaration of Robert Lochhead, Ph.D. (Ex. 2011) to support its contentions.

G. *Asserted Grounds of Unpatentability*

Petitioner asserts that claims 1–7, 9–13, and 16–19 would have been unpatentable on the following grounds:

Ground	Claim(s) Challenged	35 U.S.C. § ²	Reference(s)/Basis
1	16–17	102 or 103	Lesniak
2	1, 4–7, 9–12, 18–19	103	Lesniak, Bianchi ’518
3	2	103	Lesniak, Bianchi ’518, Phinney

² Because the ’647 patent has an earliest possible effective filing date after March 16, 2013, it is subject to the post-grant review provisions of the America Invents Act. *See* The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (“AIA”) § (3)(n)(1)(A) (2011).

Ground	Claim(s) Challenged	35 U.S.C. § ²	Reference(s)/Basis
4	3, 13	103	Lesniak, Bianchi '518, Phinney, Nabavi
5	16, 17	103	Native, Bianchi '254
6	1, 2, 4–7, 9–12, 18, 19	103	Native, Bianchi '254, Easy Homemade
7	3, 13	103	Native, Bianchi '254, Easy Homemade, Nabavi

H. Claim Construction

The challenged claims “should be read in light of the Specification, as it would be interpreted by one of ordinary skill in the art.” *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1260 (Fed. Cir. 2010). Thus, we generally give claim terms their ordinary and customary meaning. *See In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007) (“The ordinary and customary meaning is the meaning that the term would have to a person of ordinary skill in the art in question.” (internal quotation marks omitted)); *see also* 37 C.F.R. § 42.100(b) (stating that claims are construed in IPRs according to the same standard as used in federal court).

Petitioner does not argue that any claim terms require express construction. *See generally* Pet.

Patent Owner asserts that the preamble of independent claim 1 reciting “a deodorant stick” is limiting” as it provides antecedent basis for certain other claim elements, specifically, for the recitation of “said deodorant stick being free of an aluminum salt” and “said deodorant stick” having a particular hardness range. PO Resp. 8–9. Patent Owner also points to dependent claims 5–8 as relying on the preamble reciting “a deodorant stick” as antecedent basis for claim elements. *Id.* at 10. We agree.

Petitioner does not provide argument to the contrary. See, e.g., Pet. 24 (arguing that Lesniak discloses a deodorant stick in the form of an antiperspirant stick).

We determine that no express construction of any claim term is necessary to determine whether to institute *inter partes* review. *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (“[W]e need only construe terms ‘that are in controversy, and only to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))). To the extent further discussion of the meaning any claim term is necessary to our decision, we provide that discussion below in our analysis of the asserted grounds of unpatentability.

I. Level of Ordinary Skill in the Art

Petitioner proposes that a person of ordinary skill in the art (“POSA” or “POSITA”) at the time of the invention “would have (1) at least a B.S. or equivalent degree; and (2) at least two years of experience in cosmetics formulations.” Pet. 24 (citing Ex. 1002 ¶¶ 27–28). Patent Owner does not dispute Petitioner’s proposal about the POSA’s qualifications. See generally PO Resp.

We adopt and apply Petitioner’s proposed definition of a POSA, which is consistent with the level of skill reflected in the asserted prior art.

II. ANALYSIS

A. Introduction

“In an *inter partes* review, the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016)

(citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim”)). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015). Moreover, a petitioner should not “place the burden on [the Board] to sift through information presented by the Petitioners, determine where each element [of the challenged claims] is found in [the cited references], and identify any differences between the claimed subject matter and the teachings of [the cited references.]” *Google Inc. v. EveryMD.com LLC*, IPR2014-00347, Paper 9 at 25 (PTAB May 22, 2014).

1. Anticipation

To show anticipation under § 102, each and every claim element, arranged as in the claim, must be found in a single prior art reference. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359 (Fed. Cir. 2008). The prior art need not, however, use the same words as the claims in order to find anticipation. *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009). It is also permissible to take into account not only the literal teachings of the prior art reference, but also the inferences an ordinarily skilled person would draw from the reference. *Eli Lilly and Co. v. Los Angeles Biomedical Res. Inst. at Harbor-UCLA Med. Ctr.*, 849 F.3d 1073, 1074–75 (Fed. Cir. 2017); *In re Preda*, 401 F.2d 825, 826 (CCPA 1968). A reference may also anticipate a claim even if it does not expressly teach all the limitations arranged or combined as in the claim, “if a person of skill in the art, reading the reference, would ‘at once envisage’ the claimed arrangement or combination.” *Microsoft Corp. v. Biscotti, Inc.*, 878 F.3d 1052, 1068 (Fed. Cir. 2017) (quoting *Kennametal, Inc. v. Ingersoll Cutting Tool Co.*, 780 F.3d

1376, 1381 (Fed. Cir. 2015)).

2. *Obviousness*

A claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. *See KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness, if any. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In analyzing the obviousness of a combination of prior art elements, it can be important to identify a reason that would have prompted one of skill in the art “to combine . . . known elements in the fashion claimed by the patent at issue.” *KSR*, 550 U.S. at 418. A precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *Id.* Rather, “any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed.” *Id.* at 420.

Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (internal

quotation marks omitted). “Both the suggestion and the expectation of success must be founded in the prior art, not in the applicant’s disclosure.” *In re Dow Chemical Co.*, 837 F.2d 469, 473 (Fed. Cir. 1988).

An obviousness analysis “need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.” *KSR*, 550 U.S. at 418; *see In re Translogic Tech, Inc.*, 504 F.3d 1249, 1259 (Fed. Cir. 2007). In *KSR*, the Supreme Court also stated that an invention may be found obvious if trying a course of conduct would have been obvious to a POSITA:

When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense. In that instance the fact that a combination was obvious to try might show that it was obvious under § 103.

550 U.S. at 421. “*KSR* affirmed the logical inverse of this statement by stating that § 103 bars patentability unless ‘the improvement is more than the predictable use of prior art elements according to their established functions.’” *In re Kubin*, 561 F.3d 1351, 1359–60 (Fed. Cir. 2009) (citing *KSR*, 550 U.S. at 417).

We analyze the asserted grounds of unpatentability in accordance with these principles to determine whether Petitioner has met its burden to show by a preponderance of the evidence that the challenged claims are unpatentable.

B. Summary of the Cited Prior Art

1. Lesniak (Ex. 1003)

Lesniak is titled “NON-GREASY PERSONAL CARE COMPOSITIONS.” Ex. 1003, code (54). Lesniak provides an anhydrous personal care composition, comprising

(a) at least one ester, wherein said at least one ester is selected from one or more of the following groups: (i) triglycerides, (ii) diglycerides, (iii) monoglycerides, (iv) monoesters of dials, and (v) diesters of dials, and wherein a total concentration of said at least one ester is 45 to 60 weight %, based on the total weight of the composition; (b) a first wax having a melting point of 35° C. to 72° C. present in an amount of 5 to 25 weight %, based on the total weight of the composition; (c) a second wax having a melting point of 73° C. to 90° C. present in an amount of 0.5 to 15 weight% based on the total weight of the composition; (d) at least one plant oil having a saponification value of 150 to 275 mg KOH/g, wherein a total concentration of said at least one plant oil is from 5 to 35 weight%, based on the total weight of the composition.

Id. at 5:32–48. Lesniak describes that “[o]ptionally, the at least one ester is a combination of caprylic triglyceride and capric triglyceride” in ranges from 50–60 weight %, based on the total weight of the composition. *Id.* at 2:38–39, 3:42–44. “Optionally, the composition further comprises zinc oxide . . . present in a concentration of 10 to 30 weight %.” *Id.* at 3:39–42. Lesniak describes that the composition may be in the form of a stick. *Id.* at 4:22. According to Lesniak, the compositions possess “good micro-robustness, . . . reduced greasiness and an improved after-feel (such as reduced tackiness and drag) on the skin as compared to other anhydrous compositions” and “are also physically stable at high and low temperatures.” *Id.* at 5:50–54.

Lesniak discloses a desire to “to formulate such compositions from all-natural ingredients (i.e. ‘100% natural’ compositions).” *Id.* at 5:29–31.

Lesniak discloses that “the term ‘natural compositions’ indicates that the composition is formulated from ingredients which are extracted from the earth or sea, derived from plants, or synthesized by other organisms.” *Id.* at 4:61–64. Lesniak discloses that the composition may comprise “essential oils which may be used as natural fragrances.” *Id.* at 8:38.

2. *Phinney (Ex. 1004)*

Phinney is titled “DEODORANT FORMULATION.” Ex. 1004, code (54). Phinney relates to a deodorant “formulation and preparation thereof with natural ingredients in the absence of heavy metal complexes and/or petroleum based or related compounds.” *Id.* at 1:7–9. Phinney discloses that “a mixture of the hydroxides of zinc and magnesium could provide long lasting protection against odour due to perspiration . . . and are dispersible in aerosols, moisturizing creams, sticks, sprays and roll-on type applicators.” *Id.* at 1:17–22.

Phinney explains that zinc and aluminum oxide are “very unreactive against perspiration odour due to insolubility and low hydrolysis rate.” *Id.* at 2:35–36. According to Phinney, “magnesium hydroxide, . . . has low solubility and therefore does not ‘wash away’ under heavy perspiration” and thus “provides 16–20 hour protection and more for a broad cross section of people.” *Id.* at 5:47–51. Phinney discloses an embodiment of a specific formulation prepared as a solid stick that “includes magnesium oxide in an amount of between 1% and 2.5% by weight and zinc oxide in an amount of between 0.5% and 1.5% by weight.” *Id.* at 6:32–34, 6:12, 8:46–57 (claim 1).

3. *Bianchi '518 (Ex. 1006)*

Bianchi '518 is titled “LOW RESIDUE SOLID ANTIPERSPIRANT.” Ex. 1006, code (54). Bianchi '518 “relates to antiperspirant sticks, substantially free of water and long chain fatty alcohols, which provide the user with excellent antiperspirant efficacy, reduced residue when the composition is first applied to the skin, reduced residue after dry down, high temperature stability, and excellent cosmetics and aesthetics. *Id.* at 1:8–13.

Bianchi '518 discloses that the antiperspirant composition possesses a specification of “Penetration-ASTM D-1321@ 100/77/5 10–16.” *Id.* at 2:57–58. Bianchi '518 explains that

[o]ne important physical parameter affecting consumer acceptance of suspensoid antiperspirant sticks of the invention is hardness as measured by needle penetration. Typical compositions of the invention will range in hardness from about 7.0 to about 14.0 mm as shown by the above mentioned needle penetration test. Products outside this range will typically generate low consumer acceptance ratings.

Id. at 6:23–29.

4. *Native (Ex. 1007)*

Native is a printed publication from a website advertising a deodorant product sold in 2016. Ex. 1007, 5–8, 11–12, 14–18. Native is advertised as a “Deodorant That Isn’t A Chemistry Experiment,” which contains “NATURAL INGREDIENTS” and is “ALUMINUM-FREE.” *Id.* at 5. Native lists each ingredient for the Native deodorant stick product, including, *inter alia*, shea butter, beeswax, baking soda, and stearyl alcohol. *Id.* at 6, 11.

The picture on page 5 of Native, reproduced below, illustrates a deodorant product. *Id.* at 5.



The picture of the Native deodorant product, reproduced above, illustrates the deodorant stick in a supine position with the cap removed. *Id.*

5. *Bianchi '254 (Ex. 1008)*

Bianchi '254 is titled “ANTIPERSPIRANT STICK COMPOSITIONS.” Ex. 1008, code (54). Bianchi '254 discloses “antiperspirant stick compositions comprising specified amounts of antiperspirant active; carrier oil comprising volatile silicone oil and non-volatile masking oil; structurant comprising fatty alcohol and cosmetically acceptable wax having a melting point of 75 to 95° C., a portion of wax comprises polyethylene in specified amounts.” *Id.* at code (57). Bianchi '254 addresses

a need for non-whitening fatty alcohol-containing antiperspirant sticks, particularly sticks containing fatty alcohol at levels of 12% by weight or above, more particularly, 15 wt % or above. Also desired are non-whitening fatty alcohol-containing sticks that have desirable antiperspirant payouts, while providing a ‘clean’, non-sticky feel. There also remains a need for a non-whitening fatty-alcohol containing stick that provides desirable payout as well as being thermally stable and resistant to dome collapse at temperatures of 50° C.

Id. ¶ 10. Table 5 of Bianchi '254 provides eight different formulations of antiperspirant sticks with each of their respective components and amounts listed. *Id.* ¶ 131.

Bianchi '254 explains that the “[t]he hardness of the sticks can be measured in a needle penetration test.” *Id.* ¶ 26. Bianchi '254 describes the method for measuring stick hardness using a “brass/steel tapered needle (DIN 51-579)” depth on six locations of the stick, “with the hardness values being reported as an average of the six measurements.” *Id.* ¶ 117.

6. *Easy Homemade (Ex. 1009)*

Easy Homemade is a printed publication of a blog post titled “Easy Homemade Deodorant Recipe with just 3-Ingredients.” Ex. 1009, 5. Easy Homemade describes the blogger’s first experience with natural deodorant, but falling “victim to the burning fiery rash you get after using deodorant with baking soda.” *Id.* at 6. Easy Homemade describes that the blogger discovered that she “could swap the baking soda with magnesium hydroxide - basically the same thing as Milk of Magnesia - and it worked just as well.” *Id.*

The Easy Homemade blogger describes melting “two natural deodorants” that are aluminum-free, and making a “[h]uge difference” in each deodorants’ performance by adding “arrowroot powder and magnesium hydroxide to them.” *Id.* at 7. Easy Homemade provides a deodorant recipe and lists “2.15 oz. natural aluminium free deodorant (base) of choice[,] 1 oz. arrowroot powder[,] .5 oz. magnesium hydroxide[,]” to which one may “[a]dd essential oils if desired.” *Id.* at 7–8.

7. *Nabavi (Ex. 1067)*

Nabavi, titled “Antibacterial Effects of Cinnamon: From Farm to Food, Cosmetic and Pharmaceutical Industries,” is a review of the literature about the antibacterial effects of cinnamon. Ex. 1067, 7729. The review concluded that “[c]innamon use in food products and cosmetics could be a good strategy to reduce or avoid bacterial degradation and thus to reduce the incidence of infection caused by food and cosmetics.” *Id.* at 7742.

C. *The Lesniak Grounds (Grounds 1–4)*

1. *Ground 1 (Claims 16 and 17)*

Petitioner asserts that claims 16–17 are either anticipated by or would have been obvious over Lesniak. Pet. 6, 24–30. In support of its assertion, Petitioner provides a detailed discussion and claim chart for this ground explaining how each claim limitation is disclosed in Lesniak and how its assertions are supported by the testimony of Dr. Michniak-Kohn. *See* Pet. 26–30 (Ground 1; Ex. 1002 ¶¶ 56–75).

Patent Owner disagrees and provides several arguments for why Lesniak does not anticipate or render the challenged claims obvious. PO Resp. 16–33; Sur-reply 17–22.

After considering all of the arguments and cited evidence of record, we determine that the preponderance of the evidence supports the conclusion that Lesniak anticipates each of challenged claims 16 and 17. We address the parties’ arguments below. We begin our analysis addressing the disputed elements of claim 16 and 17.

a) *Independent claim 16 and Dependent Claim 17*

Claim 16 recites a deodorant stick comprising at least 60% by weight liquid triglyceride, at least one antimicrobial, and a fragrance composition of

at least about 50% by weight of natural oils, essential oils, or a combination thereof. Ex. 1001, 28:34–40. Claim 16 further requires that the deodorant stick be free of an aluminum salt. *Id.* at 28:41.

Petitioner notes that Lesniak teaches personal care compositions formed from natural ingredients, including a deodorant stick in the form of an antiperspirant stick. Pet. 24 (citing Ex. 1003, 1:16–35, 4:22, 4:61–67, 5:5–32, 10:27, 101–33), 26 (citing Ex. 1003, 4:22, 9:45, 10:27; Ex. 1002 ¶ 61). Petitioner also asserts that Lesniak teaches compositions that may include up to 60% of a liquid triglyceride such as caprylic/capric triglyceride, a primary antimicrobial (i.e., zinc oxide), *see* Ex. 1002 ¶ 62, a fragrance composition of at least 50% essential oils (in particular, 100%), and a primary structurant with a melting point of at least 50 °C (i.e., beeswax). *Id.* at 26–30 (claim chart). Petitioner also notes that Lesniak’s compositions do not contain aluminum salts or synthetic fragrances. Pet. 25 (citing Ex. 1003, 3:50–4:21, 8:37–42).

Patent Owner responds to Petitioner’s challenge with several arguments, which we address below. PO Resp. 16–33. Patent Owner summarizes its position as follows.

In sum, Lesniak provides no teaching of a stable “deodorant stick,” no teaching of a stable “antiperspirant stick,” and at least nine different sunscreen formulas with slight variations that are all unstable. Among all of this is one example *sunscreen* “stick” as to which stability is unknown. A POSA armed with Lesniak’s disclosure on the fragility of its crystal-network-structured compositions—which lacks guidance as to how to formulate a stable deodorant or antiperspirant stick—would not have been enabled to make and use the claimed “deodorant stick.” None of Petitioner’s other references remedy this deficiency; Petitioner does not attempt to argue that they do.

Id. at 31 (citing Ex. 2011 ¶¶ 55–69).

b) Whether Lesniak discloses a deodorant stick

Patent Owner first contends that “Lesniak does not disclose a ‘deodorant stick,’ as required by the preamble of the independent claims.” PO Resp. 16. Patent Owner acknowledges that Lesniak suggests that “[t]he composition can also be an antiperspirant” (Ex. 1003, 10:27–29), but contends that Lesniak fails to “specify any form of antiperspirant, which can come in a variety of forms, including aerosol sprays, creams, liquids, and gels.” PO Resp. 20–21. Patent Owner further contends that Lesniak’s “passing references to ‘antiperspirant’ fail to convert Lesniak into something other than a reference about the embodied sunscreens and diaper creams.” *Id.* at 21.

We are not persuaded by Patent Owner’s arguments. Rather, we agree with Petitioner that Lesniak teaches anhydrous personal care compositions where “[t]he composition can . . . be an antiperspirant” and “can also be a stick.” Ex. 1003, 9:45, 10:27. Thus, Lesniak teaches or suggests a deodorant in the form of an antiperspirant. Pet. Reply 3 (citing Ex. 1036, 2 (“all antiperspirants are deodorants”); Ex. 1037, 246 (same); Ex. 1084, 33 (“[T]he main mechanisms through which deodorant products intended to combat body odors . . . were the usage of antiperspirants that prevent perspiration[.]”)). In light of those express disclosures, we are unpersuaded by Patent Owner’s arguments that Lesniak fails to teach or suggest a “deodorant stick.”

c) Whether antiperspirants must contain aluminum

Patent Owner contends that Lesniak’s antiperspirants must contain aluminum. PO Resp. 20–21. Specifically, Patent Owner contends that,

To a POSA, the term “antiperspirant” means a composition containing an aluminum salt or complex—the only antiperspirant actives approved by the FDA. *See* 37 C.F.R. 350. As Dr. Lochhead explains, a POSA would have understood that the U.S. Food and Drug Administration regulates antiperspirants as drugs and regards deodorants as cosmetics. (Ex. 2011 at ¶ 51.) A POSA would have further understood Lesniak’s disclosure of “antiperspirant” as requiring the use of an antiperspirant active, and the only antiperspirant actives approved by the FDA all contain aluminum. (*Id.*)

PO Resp. 21 (footnote omitted); *see also* PO Sur-reply 18–19 (a POSA’s understanding of “antiperspirants,” now and at the time of Lesniak, is that they contain aluminum (citing Ex. 1081,³ 60:12–14 (“[A] POSA would understand if you’re talking about an antiperspirant, it has to have aluminum.”); Ex. 2011 ¶¶ 51–54).

We are not persuaded by Patent Owner’s argument. Rather, we agree with Petitioner that, “[w]hile [Patent Owner] emphasizes that the FDA requires antiperspirants marketed in the U.S. to contain aluminum, anticipation and obviousness are distinct from FDA regulatory requirements.” Reply 4 (citing *Persion Pharms. LLC v. Alvogen Malta Ops. Ltd.*, 945 F.3d 1184, 1193 (Fed. Cir. 2019) (“FDA’s approval requirements do not undermine the force of the evidence as to obviousness”)). Indeed, the evidence of record shows that zinc compounds were known antiperspirant actives that could be alternatives to aluminum. *Id.* at 5 (citing Ex. 1003, 3:39–4:21, 8:46–48, 10:27 (describing zinc-based antiperspirant compositions); Ex. 1014 ¶ 20 (“[A]ntiperspirant actives include astringent metallic salts, particularly inorganic and organic salts of aluminum, zirconium, and zinc, as well as mixtures thereof.”)); *see also* Ex. 1002 ¶ 76

³ Ex. 1081, Dr. Robert Lochhead Deposition Transcript, July 10, 2025.

(“Lesniak’s formulations include zinc oxide, which was a known alternative to aluminum.”) (citing Ex. 1003, 1:39–2:30, 3:43–4:21, 5:32–6:23, 8:31–9:59).

d) Whether Lesniak enables a deodorant stick

Patent Owner contends that Lesniak fails to provide sufficient guidance as to how to formulate a stable deodorant or antiperspirant stick, and therefore, is non-enabling. PO Resp. 25–31 (citing Ex. 2011 ¶¶ 55–56). In particular, Patent Owner contends that “the only ‘stick’ formulation Lesniak expressly teaches is a sunscreen stick.” *Id.* at 26 (citing Ex. 1003, Table 7). Patent Owner also directs our attention to the results disclosed in Lesniak’s Table 2 and argues that Lesniak’s compositions suffered from stability issues. *Id.* at 27–31; *see also* PO Sur-reply 17 (“Lesniak’s Table 2 shows that even small changes in a composition can yield unstable results,” demonstrating that the art is unpredictable.). Patent Owner concludes that “Lesniak provides no teaching of a stable ‘deodorant stick,’ no teaching of a stable ‘antiperspirant stick,’ and at least nine different sunscreen formulas with slight variations that are all unstable.” PO Resp. 30 (citing Ex. 2011 ¶ 68).

We find that the evidence of record fails to support Patent Owner’s argument that the Lesniak fails to enable a stable deodorant stick. Rather, we are persuaded by Petitioner that Lesniak sufficiently teaches compositions with the relevant ingredients and achieves stable compositions through the use of a second wax having a melting point of 73 to 90° C. Pet. Reply 8 (citing Ex. 1003, 1:39–54). For example, Lesniak’s compositions (nos. 8 and 11) that used such a second wax, namely CastorWax MP80, are disclosed as sufficiently stable. Ex. 1003, Table 2. Here, we further credit the testimony of Dr. Michniak-Kohn that

Lesniak teaches how to make stable compositions by including a secondary wax, such as castor wax, with a sufficiently high melting point. Ex. 1003, 1:39-54, 5:23-32, Table 2. Both of the compositions (nos. 8 and 11) tested by Lesniak that had such a wax passed the stability tests.

Ex. 1086 ¶ 7; *see also* Reply 8 (“Lesniak overcame any stability issues, as it expressly discloses compositions that are ‘physically stable at high and low temperatures.’”) (citing Ex. 1003, 5:23–32).

Having considered the parties’ positions and evidence of record, summarized above, we determine that Lesniak provides sufficient disclosure to enable a person of ordinary skill in the art to formulate a deodorant stick within the parameters of claims 16 and 17.

e) Conclusion as to Claims 16 and 17

We agree with Petitioner that “the challenged claims recite nothing more than well-known deodorant ingredients performing their conventional functions.” Pet. Reply 2. Therefore, for the reasons discussed above, we conclude that Petitioner has proved by a preponderance of evidence that Lesniak anticipates claims 16 and 17.

2. Ground 2 (Claims 1, 4–7, 9–12, 18, and 19)

Petitioner contends that claims 1, 4–7, 9–12, 18, and 19 would have been obvious over Lesniak and Bianchi ’518. Pet. 41–50. Petitioner relies on the teachings of Lesniak as discussed above for Ground 1, but recognizes that “Lesniak does not expressly teach hardness as measured by penetration with an ASTM-D 1321 needle.” Pet. 41; Ex. 1002 ¶ 76. Petitioner relies on Bianchi for this teaching. Pet. 41; Ex. 1002 ¶ 76. Petitioner asserts that “Bianchi ’518 teaches that the claimed hardness range of about 7.0 mm to about 14.0 mm (equivalent to about 70 mm*10 to about 140 mm*10), as measured by penetration with an ASTM-D needle, was ‘typical,’ and

products outside of the range were known to generate low consumer acceptance ratings.” Pet. 41 (citing Ex. 1006, 2:56–58, 6:23–29); Ex. 1002 ¶ 76.

Petitioner asserts a POSA would be motivated to combine the teachings of Lesniak and Bianchi ’518 because hardness was “an important physical parameter affecting consumer acceptance,” and Lesniak “is directed to formulating deodorant sticks that have the ‘skin feel’ desired by consumers.” Pet. 31–32 (citing Ex. 1002 ¶ 76; Ex. 1003, 1:16–35). Petitioner concludes that “a POSA would have looked at publications, like Bianchi ’518, to find consumer acceptable hardness ranges in formulating a stick composition acceptable to consumers,” and would have had a reasonable expectation of success in doing so because Bianchi ’518 “had successfully formulated antiperspirants to have a hardness within that range.” *Id.* at 31–32 (citing Ex. 1002 ¶¶ 77–79; Ex. 1006, 2:56–58, 6:23–29). Petitioner concludes that a POSA would have understood how to adjust hardness by altering the amount of the primary structurant and/or by incorporating secondary structurants. *Id.* at 32–33 (citing Ex. 1003, 9:45–54, 1:39–2:30, 3:43–4:21, 5:32–48; Ex. 1014 ¶¶ 3, 28–29).

Patent Owner disagrees and provides several arguments for why Lesniak does not anticipate or render the challenged claims obvious. PO Resp. 16–41; Sur-reply 17–22.

After considering all of the arguments and cited evidence of record, we determine that preponderance of the evidence supports the conclusion that Lesniak renders obvious each of challenged claims 1, 4–7, 9–12, 18, and 19. We address the parties’ arguments below. We begin our analysis addressing the disputed elements of claim 1.

a) Independent Claim 1

In addition to the arguments Patent Owner raised concerning Ground 1 that we address above, Patent Owner asserts that a POSA would have no reason to modify Lesniak and neither Lesniak nor Bianchi '518 discloses the claimed stick harness range. PO Resp. 32–41. We address each of these arguments in turn.

(1) Motivation to Combine Lesniak and Bianchi '518

Patent Owner asserts that Petitioner's argument is inconsistent because it asserts that a POSA "would have understood Lesniak's formulations *to be complete deodorant sticks that did not require additional ingredients*," see Pet. 29, 35, yet "also contends that a POSA would have sought to modify Lesniak by adding and changing a host of different ingredients and their amounts." PO Resp. 32 (citing Pet. 24–37; Ex. 1002 ¶ 76; Ex. 2012, 59:4–7). Patent Owner compares the formulations of Lesniak and Bianchi '518 asserting that a "POSA would not have been motivated to consider the teachings of Bianchi-518 in combination with Lesniak, as the Bianchi compositions require large amounts of non-natural ingredients (volatile silicones), and using such non-natural ingredients would defeat the core objective of Lesniak to provide natural compositions." PO Resp. 33–34.

Petitioner responds that "[a] complete deodorant is merely one that qualifies as a deodorant without the need for additional ingredients. It does not mean that Lesniak's formulations were perfect or that a POSA lacked any reason to improve or modify them." Pet. Reply 9–10 (citing Ex. 1081, 43:25–44:18 (asserting admission by statement that "there was a motivation to make [deodorants] better," notwithstanding that they were "complete").

We determine that Petitioner has the better position here. Therefore, we determine that this ground does not fail because a POSA would have had no motivation to modify Lesniak. A POSA would have motivation to make a formula that works better.

We also determine that Petitioner does not rely on the teachings of Bianchi '518 relating to its non-natural ingredients, such as silicones, but on its teaching concerning an acceptable hardness range to accommodate consumer preferences. *See* Pet. 30–34. To that point, Lesniak teaches the use of a first and second wax as preferred structurants, and the evidence of record supports a conclusion that a POSA would understand how to formulate a deodorant stick using the structurants taught in Lesniak. *See, e.g.,* Pet. 21–33; Ex. 1002 ¶ 42 (“For deodorant sticks, the structurant is what mixed with the carrier oil to cause the formulation to have the consistency (e.g., hardness) of a stick form. In addition to adjusting the balance between, e.g., waxes of different melting points to control hardness, a POSA would have appreciated that the amount of structurant controls the hardness of the stick.”). Therefore, Patent Owner’s argument concerning the lack of a reason to combine the references based on the natural or non-natural nature of the ingredients of Lesniak and Bianchi '518 is unpersuasive.

(2) Whether Lesniak nor Bianchi-518 Disclose a Stick of the Claimed Hardness

As noted above, claim 1 requires that the claimed deodorant stick has “a hardness from about 70 mm*10 to about 140 mm*10, as measured by penetration with ASTM D-1321 needle.” Ex. 1001, 27:31–33 (claim 1). For this element, Petitioner relies on Bianchi '518’s teaching that consumers prefer products falling within a hardness range of about 7.0 mm to about

14.0 mm (equivalent to about 70 mm*10 to about 140 mm*10), as measured by the standard ASTM-D 1321 needle penetration test. Pet. 30 (citing Ex. 1006, 2:56–58, 6:23–29).

Patent Owner contends that the standard hardness test referenced in Bianchi-518 is different from the test described in the '647 patent and contends that “the Bianchi hardness test uses a needle and shaft with *double the weight* of the needle and shaft used in the '647 Patent [100 grams vs. 50 grams], and thus its so-called same or similar hardness ranges reflect that the Bianchi compositions are actually *significantly harder* than those claimed in the '647 Patent (i.e., it requires twice the weight of the penetration needle assembly to yield similar penetration depths).” PO Resp. 35 (citing Ex. 2011 ¶¶ 74–78; Ex. 2018,⁴ 5–6 (“Place a 50-g weight above the penetrometer needle, making a total load of 100 ± 0.015 g for the needle and all attachments.”)). Patent Owner supports its argument with the testimony of Dr. Lochhead, who explains that

The '647 Patent uses a different testing method. The '647 Patent explains that its invention exhibits a hardness in the range of “70 mm*10 to about 140 mm*10,” as measured using an ASTM D-1321 penetration needle in a test similar to ASTM D-1321, but with one significant difference: “the total weight of the shaft and needle is 50 ± 0.2 grams when the shaft is in free fall.” ('647 at 23:25-26.) The '647 Patent’s method does not include the “50-g weight” added to the needle assembly as an additional load, which is used in the standard ASTM D-1321 test utilized by Bianchi-518.

Ex. 2011 ¶ 82; *see* PO Resp. 36. Patent Owner further contends that the evidence of record fails to support a conclusion that Bianchi '518’s test and

⁴ Ex. 2018, ASTM D-1321 Needle Penetration Test Standard (1996).

the '647 patent's test would produce overlapping hardness ranges. *Id.* at 37.⁵

We find that the evidence of record fails to support Patent Owner's position that the '647 patent discloses a unique hardness test. As noted by Petitioner, the '647 patent discloses that hardness was measured by "penetration with ASTM D-1321 needle (see Hardness test method below)." Ex. 1001, 15:37–40; *see also id.* at Abst., 2:19–20. The '647 patent describes that its hardness test was based on ASTM D-1321 and used penetration needles "as specified for the ASTM D 1321 D 1321/DIN 51 579, Officially certified, Taper-Tipped needle, No. H-1310, Humboldt Mfg." *Id.* at 23:12–15. The '647 patent further discloses that "[p]eriodic calibration should confirm: . . . the total weight of the shaft and needle is 50 ± 0.2 grams when the shaft is in free fall." *Id.* at 23:20–26. That disclosure for periodic calibration coincides with the requirement under the ASTM D-1321 test that "[t]he final weight of the needle shall be 2.5 ± 0.05 g. The total weight of the plunger shall be 47.5 ± 0.05 g." Ex. 2018, 5. However, the ASTM D-1321 test also makes clear that a 50-gram weight is placed "above the penetrometer needle, making a total load of 100 ± 0.15 g for the needle and all attachments," which is consistent with Bianchi '518's disclosure of a test using 100 grams of weight. PO Resp. 36; Ex. 1006, 2:58, 6:25-28.

The '647 patent, on the other hand, fails to expressly disclose whether the test was performed without the required 50-gram weight placed above

⁵ Patent Owner asserts that Petitioner improperly submitted evidence with its Reply to correlate the hardness test methods of the Bianchi references and the challenged claims. *See* Papers 22, 23. Because we do not agree on the record before us that these test methods are different, we do not rely on Petitioner's newly submitted evidence and will not further address Patent Owner's assertions.

the penetrometer needle, and its silence to that fact is insufficient to persuade us that a different test was ultimately intended or performed. Rather, we are persuaded by the following testimony of Dr. Michniak-Kohn:

Dr. Lochhead states that he believes the claimed recitation of a hardness range “as measured by penetration with ASTM D-1321 needle” should not be understood to indicate the ASTM D-1321 standard, but a modification of that standard. Ex. 2011, 37. As a basis for his opinion. Dr. Lochhead focuses on language in the ’647 Patent that “the total weight of the shaft and needle is 50 ± 0.2 grams.” Ex. 2011, ¶26. But he omits the preceding language, which states that this is the weight to be used for “[p]eriodic calibration” to “confirm” that the equipment is in proper working order. Ex. 1001, 23:20-29. That “[p]eriodic calibration” section precedes a section identifying what needs to be confirmed “[a]t the time of use.” *Id.*, 23:29-36. Those two sections would suggest to a POSA routine checking of equipment. Only after those two sections does the specification set forth the test. *Id.*, 23:37–24:14 (“Sample Preparation and Measurement”). Moreover, the “[p]eriodic calibration” comports with the ASTM D-1321 standard, which likewise requires that the “weight of the needle shall be 2.5 g” and the weight of the plunger shall be 47.5 g,” for a 50-g total weight of the needle assembly. Ex. 2018, 5. Although the ASTM D-1321 standard further describes using an additional 50-g weight during the test, a POSA would not read the ’647 Patent to exclude that weight. Rather, a POSA would consult the details of the ASTM D-1321 standard identified by the ’647 Patent to ensure compliance.

Ex. 1086 ¶ 3.

Accordingly, having considered the parties’ positions and evidence of record, summarized above, we determine that a POSA would have been motivated to formulate Lesniak’s deodorant stick to be within the claimed hardness range, which overlaps with ranges known to be acceptable to consumers as disclosed in Bianchi ’518, with a reasonable expectation of success. Pet. 31–33 (citing Ex. 1002 ¶¶ 76–80).

(3) Conclusion as to Claim 1

For the reasons discussed above, we conclude that Petitioner has proved by a preponderance of evidence that the combined teachings of Lesniak and Bianchi '518 render claim 1 obvious.

b) Claims 4–7, 9–12, 18, and 19

Petitioner argues that claims 4–7, 9–12, 18, and 19 would have been obvious over the combination of Lesniak and Bianchi '518. Pet. 30–37. Patent Owner presents the arguments discussed above regarding the limitations of claim 1, but does not present arguments or direct us to evidence against Petitioner's challenges that are specific to the limitations of claims 4–7, 9–12, 18, and 19. *See generally* PO Resp.

Upon review of Petitioner's undisputed argument, which we adopt as our own, and the evidence of record, we determine that Petitioner has demonstrated by a preponderance of the evidence that each of the limitations recited in claims 4–7, 9–12, 18, and 19 are suggested by the above-identified prior art, and that the skilled artisan would have had reason to make the combination with a reasonable expectation of success. *See, e.g.*, Pet. 30–37.

c) Summary

The preponderance of the evidence supports Petitioner's argument that claims 1, 4–7, 9–12, 18, and 19 would have been obvious over the combination of Lesniak and Bianchi '518. Patent Owner does not persuade us otherwise. Accordingly, we determine that claims 1, 4–7, 9–12, 18, and 19 are rendered obvious by the combination of Lesniak and Bianchi '518.

3. Ground 3 (Claim 2)

Petitioner contends that claim 2 would have been obvious over Lesniak, Bianchi '518, and Phinney. Pet. 44–45. Claim 2 depends from

claim 1 and further requires that the primary antimicrobial be selected from a group of antimicrobials, one of which is magnesium hydroxide. *See* Ex. 1001, 27:34–40. Petitioner relies on Phinney for teaching this additional limitation of using magnesium hydroxide in addition to zinc oxide as an antimicrobial in a deodorant stick. *See* Pet. 37.

Petitioner provides a detailed claim chart for this ground explaining how each claim limitation is disclosed in the asserted references and how its assertions are supported by the testimony of Dr. Michniak-Kohn. *See* Pet. 33–37; Ex. 1002 ¶¶ 102–109.

Patent Owner contends that Lesniak and Phinney describe entirely different uses for their metal oxides/hydroxides. PO Resp. 42. In particular, Patent Owner contends that Lesniak includes zinc oxide in sunscreen and diaper cream compositions as a protective barrier, not for odor control. *Id.* at 42–43. Moreover, Patent Owner contends that “Lesniak’s formulas with high concentrations of zinc oxide [are] unsuitable for use as an underarm deodorant or antiperspirant.” *Id.* (citing Ex. 2011 ¶ 26; Ex. 1004, 2:33–36 (“[Z]inc oxide... [is] very unreactive against perspiration odour.”), 3:20–21). Thus, a “POSA would not have been motivated to look at Phinney to modify Lesniak or incorporate any additional antimicrobial into Lesniak ‘to improve odor protection,’ as odor protection was not a goal of Lesniak’s zinc oxide in the first place.” *Id.* at 43.

Furthermore, Patent Owner contends that Phinney does not suggest a combination of zinc oxide and magnesium hydroxide. PO Resp. 43–44. Rather, according to Patent Owner,

Petitioner cherry picks the single ingredient magnesium hydroxide from Phinney’s disclosure of a three- or four-ingredient combination to combat odor. Phinney explains that to combat odor, its compositions contain a combination of

(1) magnesium hydroxide, *and* (2) zinc hydroxide, *and* (3) potassium chloride and/or potassium sulfate. (Ex. 1004 at 8:7-31, cl. 1.) There are two significant aspects to this multi-part odor fighting combination: the requirement of zinc hydroxide, not zinc oxide; and the requirement of potassium chloride/sulfate. (Ex. 2011 at ¶ 67.)

Id. at 43. Patent Owner further contends that Lesniak and Phinney are incompatible, arguing that Lesniak's compositions are anhydrous while Phinney's compositions require water. *Id.* at 46 (citing 2011 ¶¶ 100); *see also* Ex. 1004, 4:48–51, 8:22–41 (explaining magnesium oxide and zinc oxide are both mixed with water to become their respective hydroxides, and that they “must be completely converted to their hydroxides or skin irritation will occur”); Sur-reply 20–21.

We have considered Patent Owner's arguments but find, on this record, Petitioner to have the better position. In particular, we are persuaded that a POSA would have been motivated to add magnesium hydroxide to Lesniak's compositions because

Phinney teaches that: (i) zinc oxide has “low activity” as a “deodorant agent”; (ii) magnesium hydroxide is “an effective deodorant”; and (iii) zinc and magnesium compounds have synergistic effects, hence motivating a POSA to add magnesium hydroxide to Lesniak's zinc oxide deodorants. Ex. 1004, 1:13–3:16, 5:61–67.

Pet. Reply 10; *see also* Pet. 27–38 (arguing that a POSA would have found it obvious to add magnesium hydroxide to the composition of Lesniak because zinc oxide is not a particularly efficacious antimicrobial in deodorants when used alone, whereas magnesium hydroxide is an effective deodorant).

We acknowledge that Phinney suggests mixing zinc oxide with water in order to convert zinc oxide to zinc hydroxide. *See, e.g.*, Ex. 1004, 1:17–19 (“It was also found that a mixture of the hydroxides of zinc and

magnesium could provide long lasting protection against odour due to perspiration.”). The evidence of record, however, shows that both magnesium hydroxide and zinc oxide were known options for deodorant antimicrobials, and claim 2 does not require a minimal level of effectiveness such that the combination of magnesium hydroxide and zinc oxide would need to outperform any other combination including the mixture of zinc and magnesium hydroxides. *See In re Fulton*, 391 F.3d 1195, 1200 (Fed. Cir. 2004) (“Our case law does not require that a particular combination must be the preferred, or the most desirable, combination[.]”). Here, we also credit the testimony of Dr. Michniak-Kohn that a “POSA would understand that the water-based vs. anhydrous distinction is not relevant, as a POSA would understand that magnesium hydroxide would still function as an antimicrobial in anhydrous compositions such as those disclosed by Lesniak.” Ex. 1086 ¶ 9 (citing Ex. 1072, 1–2, 5 (disclosing deodorants without added water and with zinc oxide and/or magnesium hydroxide)). Accordingly, we are persuaded by Petitioner that “a POSA would have been motivated to incorporate magnesium hydroxide into Lesniak’s composition following Phinney’s guidance, as this combination would improve odor protection while maintaining the ‘natural’ composition intended by Lesniak.” Pet. 51–52.

Ultimately, however, it does not matter whether Phinney teaches a composition that contains water. “[I]t is not necessary that [two pieces of prior art] be physically combinable to render obvious” the claimed subject matter. *Allied Erecting and Dismantling Co., Inc. v. Genesis Attachments, LLC*, 825 F.3d 1373, 1381 (Fed. Cir. 2016) (quoting *In re Sneed*, 710 F.2d 1544, 1550 (Fed. Cir. 1983)). Rather, the question is “whether the claimed inventions are rendered obvious by the teachings of the prior art as a whole.”

In re Etter, 756 F.2d 852, 859 (Fed. Cir. 1985) (noting that whether one prior art reference can be incorporated into another is “basically irrelevant.”) Lesniak and Phinney undisputedly teach that magnesium hydroxide and zinc oxide were known ingredients in personal care products including deodorant sticks. The inclusion of magnesium hydroxide and zinc oxide into a single product therefore represents, absent evidence to the contrary, “a combination of familiar elements according to known methods that does no more than yield predictable results.” *Agrizap, Inc. v. Woodstream Corp.*, 520 F.3d 1337, 1344 (Fed. Cir. 2008).

Additionally, Patent Owner contends that Lesniak’s compositions contain large amounts of zinc oxide (up to 20%) and that a POSA would have understood that “Lesniak’s upwards of 20% zinc oxide compositions to be far from the 0.5-1.5% of zinc hydroxide used in Phinney.” PO Resp. 45–46 (citing Ex. 2011 ¶ 99; Ex. 2012, 84:14–16). We are not persuaded. The evidence of record does not support a conclusion that compositions comprising 10 to 30% zinc oxide such as disclosed in Lesniak would not be appropriate for deodorants. Again, we credit the testimony of Dr. Michniak-Kohn “that deodorants were known to contain ranges of zinc oxide that encompass the amount of zinc oxide in Lesniak’s compositions.” Ex. 1086 ¶ 9 (citing Ex. 1001, 10:47–59; Ex. 1072, 1–2 (disclosing deodorants having “from 0.5 to 20%” zinc oxide)).

For the reasons discussed above, we determine that Petitioner has shown by a preponderance of the evidence that claim 2 would have been obvious over Lesniak, Bianchi ’518, and Phinney.

4. Ground 4 (Claims 3 and 13)

Petitioner contends that claims 3 and 13 would have been obvious over Lesniak, Bianchi ’518, Phinney, and Nabavi. Pet. 6. Claim 3 depends

from claim 1 and adds the requirement of an additional microbial that may be cinnamon essential oil. Ex. 1001, 27:41–42. Claim 13 depends from claim 3 and add a water solubility requirement for the antimicrobials. *Id.* at 28:20–23. Petitioner relies on Nabavi for teaching the use of cinnamon essential oil for both its aromatic and antibacterial effects. *See* Pet. 41.

Petitioner provides a detailed claim chart for this ground explaining how each claim limitation is disclosed in the asserted references and how its assertions are supported by the testimony of Dr. Michniak-Kohn. *See* Pet. 41–45; Ex. 1002 ¶¶ 110–118.

Patent Owner asserts that “Nabavi does not disclose cinnamon essential oil either as an antimicrobial to fight odor-causing bacteria, in a deodorant stick or antiperspirant stick, or as an additional antimicrobial paired with magnesium hydroxide and/or zinc oxide. In fact, Nabavi warns against the use of cinnamon essential oil in cosmetic topical formulations due to ‘contact dermatitis’ and ‘inflammation and alteration of the surface epidermis’ after use, and therefore states ‘long standing excessive use is not recommended.’” PO Resp. 48–48 (citing Ex. 1067, 7742; Ex. 2011 ¶ 107).

We do not agree with Patent Owner’s position that a POSA would not have had a reason to combine Lesniak and Nabavi because Nabavi does not teach the use of cinnamon as an antimicrobial in an underarm deodorant. PO Resp. 48. Nabavi notes that with a “growing demand for preservative-free cosmetics, herbal extracts with antimicrobial activity have recently been used in the cosmetic industry to reduce the risk of allergies connected to the presence of methylparabens.” Ex. 1067, 7729 (Abst.). Nabavi further provides examples of cinnamon applications in the food and cosmetic industries, including studying the protective effects of cinnamon in food matrices, cosmetic products, and active packaging. *Id.* at 7738. Nabavi

concludes that “[c]innamon use in food products and cosmetics could be a good strategy to reduce or avoid bacterial degradation and thus to reduce the incidence of infection cause by food and cosmetics.” *Id.* at 7742. Nabavi shows that cinnamon may be used as an antibacterial in a broad range of products including cosmetics, and we determine on the record before us that a POSA would have looked to Nabavi’s teaching of the use of cinnamon as an antibacterial and fragrance to apply in Lesniak’s composition. *See* Ex. 1002 ¶ 112 (discussing Nabavi’s description of the use of cinnamon as an antibacterial in cosmetics and in aromatherapy).

For instance, Dr. Michniak-Kohn testifies that

A POSA would have looked at scientific publications, like Nabavi, in evaluating options for antimicrobials in deodorants. And a POSA would have been motivated to incorporate Nabavi’s cinnamon essential oil into Lesniak’s deodorant stick to enhance its aromatic and antibacterial effects in combatting odor and preventing product degradation, and because cinnamon essential oil would serve multiple purposes. Not only was it a known antimicrobial, but it was also a known fragrance. Moreover, Lesniak calls for essential oils to be used in its composition. Ex. 1003, 8:31-34.

Ex. 1002 ¶ 113. We credit Dr. Michniak-Kohn’s testimony here as supported by the record before us at this stage of the proceeding. We also agree with Petitioner that even if cinnamon essential oil was thought to cause irritation with excessive use, less-preferred combinations can still provide a motivation. *See* Pet. Reply 18 (citations omitted).

On the record before us, we determine that Petitioner has shown sufficiently a POSA would have had a reason to combine the teachings of Lesniak, Bianchi ’518, Phinney, and Nabavi.

5. *Conclusion*

For the foregoing reasons, we determine that Petitioner has shown a by a preponderance of the evidence that claims 1–7, 9–13, and 16–19 of the ’647 patent are unpatentable.

D. *The Native Grounds (Grounds 5–7)*

Petitioner presents three grounds that rely on Native to teach the majority of the limitations of the asserted claims in Grounds 5 through 7. *See* Pet. 41–66. For Ground 5, Petitioner asserts that claims 16 and 17 of the ’647 patent would have been obvious over Native and Bianchi ’254. Pet. 45–53. For Ground 6, Petitioner asserts that claims 1, 2, 4–7, 9–12, 18, and 19 would have been obvious over Native in view of Bianchi ’254 and Easy Homemade. Pet. 53–63. For Ground 7, Petitioner asserts that claims 3 and 13 would have been obvious over Native, Bianchi ’254, Easy Homemade, and Nabavi. Pet. 63–66.

Petitioner provides detailed claim charts for each of these grounds explaining how each claim limitation is disclosed in the asserted references and how its assertions are supported by the testimony of Dr. Michniak-Kohn. *See* Pet. 45–53 (Ground 5), 53–63 (Ground 6), Pet. 53–63 (Ground 7); Ex. 1002 ¶¶ 119–188.

1. *Ground 5 – Native and Bianchi ’254*

a) *Petitioner’s Contentions*

Petitioner states that “Native discloses features of deodorant stick compositions claimed in the ’647 Patent” Pet. 46. Specifically, Petitioner states:

Native sets forth the ingredients in a natural, aluminum-free deodorant stick that was being marketed and sold in at least 2016. The deodorant stick composition is comprised of

caprylic-capric triglyceride, baking soda, stearyl alcohol, hydrogenated castor oil, polyglycerol-3 beeswax, and jojoba esters. Aluminum salts, and synthetic fragrances are not included in the composition. As a POSA would have understood, Native lists its deodorant stick ingredients in order from greatest to least concentration; however, it does not expressly teach the percentage of each ingredient used, including the percentage of liquid triglyceride (i.e., carrier oil). In the absence of express teachings on these points, a POSA would have looked to other publications in the field with suitable weight percentage examples[, like Bianchi '254].

Pet. 46–47; *see also id.* at 49–63 (claim chart showing where limitations of claims 16 and 17 are taught in Native and Bianchi '254).

Petitioner relies on Bianchi '254 to teach the appropriate concentrations of the ingredients of the deodorant stick such as the percentages of liquid triglyceride, a carrier oil. *See id.* at 46–47. In providing support for the teaching of the limitation “at least 60% by weight liquid triglyceride” in claim 16, Petitioner provides the following explanation from the teachings of Bianchi '254.

Bianchi '254 describes an antiperspirant stick with 40–80% oil. In Bianchi '254, part of the oil carrier (20–40% of the total) is a natural oil such as a liquid triglyceride. Bianchi '254 also describes that the antiperspirant stick further comprised 20–40% of silicone oil. A POSA would have understood that silicone oil and liquid triglyceride both functioned as “carrier oils” that, among other things, solubilize and disperse other ingredients, and therefore would have been motivated to use even more liquid triglyceride (up to 80%) when formulating a natural deodorant stick (like in Native) that did not include, e.g., 20–40% silicone oil, to meet what the '647 Patent admits was an existing consumer preference for “more natural products.”

Thus, a POSA would have been motivated to use at least about 60% of a liquid triglyceride in a deodorant stick, and would have had a reasonable expectation of success in doing so,

inasmuch as (i) Bianchi '254 already suggests a 40% amount, and (ii) a POSA would have, based on Bianchi '254's overall oil teachings, used even more of the liquid triglyceride (up to 80%) when making a stick with only that oil carrier (as taught by Native).

The overlap between the claimed liquid triglyceride range and the ranges in the prior art renders the claim presumptively obvious. The '647 Patent discloses nothing critical or unexpected regarding the claimed range relative to the prior art ranges.

Pet. 50 (citing Ex. 1001, 25–26; Ex. 1002 ¶¶ 128–132; Ex. 1008 ¶¶ 15–17, 23, 50, claims 17 and 18).

As for claim 17's further requirement that the deodorant stick of claim 16 further comprise "a primary structurant having a melting point from about 50°C. to about 80°C.," Petitioner relies on Native's teaching of using stearyl alcohol that has the claimed melting point range as the primary structurant in Native's deodorant stick because it is listed first in the ingredient list. *See* Pet. 52 (citing Ex. 1001, 15:51–54; Ex. 1008, Tables 1, 5). Petitioner strengthens this teaching by relying further on Bianchi '254's teaching that:

Singles out stearyl alcohol as a "widely used" structurant of particular interest" for antiperspirants. Bianchi '254 describes antiperspirant formulations comprising 1-25% of a fatty alcohol, such as stearyl alcohol, as the structurant present in the greatest amount.

Thus a POSA would have been motivated to use a primary structurant with a melting point of from about 50°C. to about 80°C., such as stearyl alcohol, in a deodorant stick, inasmuch as both Native and Bianchi '254 describe doing so.

Pet. 52–53 (citing Ex. 1008 ¶¶ 1, 18, 19, 58, 131 (showing formulations with 18% stearyl alcohol as primary structurant and 3.5% castor wax as secondary structurant)).

b) Patent Owner's Contentions

Patent Owner provides three reasons why the above-described combination of Native and Bianchi '254 fails for Ground 5. First, Patent Owner asserts that Native's deodorant stick was "a known leader in the field," and the Native reference shows nothing but positive reviews for Native's deodorant stick. PO Resp. 50–51. Consequently, Patent Owner asserts, "[w]ith this level of consumer acceptance and praise, and no identified issues or concerns, there was simply no reason for a POSA to have sought to modify Native." *Id.* at 51 (citing Ex.2011 ¶¶ 110–112). Any such modification, Patent Owner asserts, is impermissibly based on hindsight using the '647 patent as a roadmap. *Id.*

Second, Patent Owner points to the differences in the Native's aluminum-free product containing natural ingredients and Bianchi '254's formulation containing volatile silicon and aluminum. PO Resp. 51–52. Patent Owner concludes that a "POSA would not have been motivated to consider the teachings of Bianchi-254 in combination with Native, as Bianchi-254's prominent use of non-natural ingredients and aluminum would defeat the core objective of Native to provide aluminum-free, natural deodorant products." *Id.* at 52.

Third, Patent Owner again points to the differences between the Native and Bianchi '254 formulations concluding:

The Petition does not explain what would have motivated a POSA to have implemented the weight-percentages from a non-natural antiperspirant stick into the natural, aluminum-free stick of Native (if it is even possible), or what a POSA would have substituted for aluminum and volatile silicones in order to preserve the same weight percentages without the presence of the aluminum active and silicones.

PO Resp. 53 (citing Ex. 2011 ¶¶ 115–118).

c) Party Responses

Petitioner responds that it is not necessary that Native itself identify any deficiencies for a POSA to be motivated to modify it, and also points to Easy Homemade as providing explicit motivation to modify Native. *See* Reply 19 (citing *KSR Int'l*, 550 U.S. 398, 418–21). Petitioner also responds that the differences between the formulations of Native and Bianchi '254 would not prevent a POSA from combining the teachings because “Native already achieved a silicone and aluminum-free deodorant stick, so a POSA therefore would not have to draw on Bianchi-254’s teachings for those features. Ex. 1002, 67-71. Native further discloses the relative amounts of each ingredient in that stick. *Id.* Bianchi-254 simply provides a POSA with guidance on the quantitative amounts of those ingredients and preferred hardness.” Pet. 21. Petitioner points out that the Specification of the '647 patent in fact “incorporates references disclosing aluminum-containing antiperspirants for their disclosure of ‘suitable’ structurants, emulsifiers, antimicrobials, and other ingredients.” *Id.* at 22 (citing Ex. 1001, 16:23–27, 19:42–47; Ex. 1041; Ex. 1083; Ex. 1081, 46:22–52.22).

Patent Owner reiterates that the differences in formulations between Native and Bianchi '254 indicate a POSA would need to use impermissible hindsight to arrive at the claimed inventions. PO Sur-Reply 23–24. Patent Owner explains that “Petitioner ignores that Bianchi-254’s ‘guidance’ on the quantitative amounts of the ingredients in question is influenced and scaled by the presence of over 50% aluminum and silicone in Bianchi’s sticks.” *Id.*

d) Discussion

Petitioner asserts that Native’s natural, aluminum-free deodorant stick is comprised of the following ingredients—caprylic/capric triglyceride, baking soda, stearyl alcohol, hydrogenated castor oil, polyglycerol-3

beeswax, and jojoba esters—although the percentage of each ingredient is not disclosed. Pet. 46 (citing Ex. 1007, 5, 11).⁶ Dr. Michniak-Kohn testifies that “[i]n the absence of express teachings on these points, a POSA would have looked to other publications in the field with suitable weight percentage examples.” Ex. 1002 ¶ 121; *see* Pet. 47.

Dr. Michniak-Kohn notes that Native’s primary structurant is a fatty alcohol as in Bianchi ’254’s antiperspirant stick compositions. Ex. 1002 ¶ 122. Dr. Michniak-Kohn asserts that “Bianchi ’254 singles out stearyl alcohol as a ‘widely used’ structurant ‘of particular interest’ for antiperspirants.” *Id.* ¶ 139 (citing Ex. 1008 ¶¶ 1, 58). Dr. Michniak-Kohn concludes that “[w]hen formulating a natural, aluminum-free deodorant stick based on the composition in Native, a POSA would have looked at publications describing other stick compositions, like Bianchi ’254, for information regarding the amount of ingredients.” *Id.* ¶ 123. We credit Dr. Michniak-Kohn’s testimony as consistent with the record evidence, and further find Dr. Lochhead’s testimony to the contrary unpersuasive.

First, we note that Petitioner looks to the teachings of Bianchi ’254 to inform the appropriate amount for the ingredients described in Native, not to alter Native’s formulation. *See* Pet. 46 (stating “it would be obvious to formulate a deodorant stick such as the stick disclosed by Native using ingredient concentrations as taught by Bianchi ’254”). Therefore, Patent Owner’s argument that the praise for the Native product would not incentivize a POSA to change the Native formulation misses the mark. Petitioner does not seek to change Native’s formulation, but verify the

⁶ Dr. Michniak-Kohn notes that aluminum salts and synthetic fragrances are not included in Native’s composition. *See* Ex. 1002 ¶ 121.

amount of the listed ingredients. *Id.* In any event, as Petitioner notes, a reason to arrive at a claimed invention does not have to come from a particular identification of deficiencies in Native itself; as the Supreme Court noted in *KSR*, such motivation may come from “any need or problem known in the field of endeavor at the time of invention and addressed by the patent.” 550 U.S. at 420; *see* Pet. Reply 19.

As Petitioner notes, both Native and Bianchi ’254 are in the same field as the ’647 patent—personal care compositions, including deodorants and/or antiperspirants. Pet. 48 (citing Ex. 1001, Abstract, 18:65–26:36; Ex. 1007, 5–8, 11; Ex. 1008, Abstract, ¶¶ 10–21; Ex. 1002 ¶ 126). There is nothing in the record that we have found, nor has Patent Owner pointed us to any such evidence, to distinguish natural, aluminum-free deodorants from antiperspirants containing aluminum in such a way that ingredients of one would not be used for the same purpose in the other. As the Supreme Court noted in *KSR*, “[t]he combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” 550 U.S. at 416. In fact, as Petitioner points out, Patent Owner in the ’647 patent “incorporates references disclosing aluminum-containing antiperspirants for their disclosure of ‘suitable’ structurants, emulsifiers, antimicrobials, and other ingredients.” *Id.* at 22 (citing Ex. 1001, 16:23–27, 19:42–47; Ex. 1041; Ex. 1083; Ex. 1081, 46:22–52.22).

Dr. Michniak-Kohn persuasively asserts on the record before us that when looking at particular ingredients of a formulation, a POSA would look to other references in the same field—personal care compositions, including deodorants and/or antiperspirants—to determine how much of each ingredient should be used. We are not persuaded by the testimony of Dr. Lochhead to the contrary. Dr. Lochhead testifies that:

In my opinion, a POSA seeking to develop a natural deodorant stick would not have been motivated to implement the weight percentages disclosed in Bianchi-254. Bianchi-254 describes and claims antiperspirant formulations containing 50% or more by weight of aluminum and silicone. For example, the antiperspirant compositions disclosed in Bianchi-254 contain, *e.g.*, “from 20 to 40 wt. %” of a volatile silicone, and silicone is the ingredient often present in the greatest amount across Bianchi-254’s compositions. (Ex. 1008 at [0042], Tables 1, 5; Ex. 2012 at 108:10-14.) The Bianchi-254 compositions further include from 20 to 30 wt. % of the aluminum-containing antiperspirant active. (Ex. 1008 at Tables 1, 5; Ex. 2012 at 108:10-14.)

....

Bianchi-254 does not disclose a “natural, aluminum-free deodorant stick.” The Petition does not explain what would have motivated a POSA to have implemented the weight-percentages from a non-natural antiperspirant stick into the natural, aluminum-free stick of Native (which, in my opinion, may not even be possible), or what a POSA would have substituted for aluminum in order to preserve the same weight percentages without the presence of the mixed aluminum salt. The Petition does not explain how a POSA would have had a reasonable expectation of success in implementing Bianchi-254’s teachings while maintaining a stable composition, and in my opinion they would not have a reasonable expectation of success in doing so.

Ex. 2011 ¶¶ 115, 117.

Dr. Michniak-Kohn provides a convincing rationale for a POSA to look to Bianchi-254 for weight-percentages for the ingredients of Native even though Bianchi-254 is not a “natural” deodorant. Dr. Michniak-Kohn notes that Native’s primary structurant is stearyl alcohol, a fatty alcohol, and Bianchi ’254 “is a publication in the field which describes ‘fatty-alcohol containing antiperspirant stick compositions’ and methods related thereto.”

Ex. 1002 ¶ 122.

Dr. Michniak-Kohn further persuasively testifies that the combination of Native and Bianchi '254 teaches the “at least 60% by weight liquid triglyceride” limitation. Dr. Michniak-Kohn states:

Bianchi '254 describes an antiperspirant stick with 40-80% oil. In Bianchi '254, part of the oil carrier (20-40% of the total) is a natural oil such as a liquid triglyceride. Ex. 1008, ¶¶[0017], [0050]. Bianchi '254 also describes that the antiperspirant stick further comprised 20-40% of silicone oil. Id., ¶[0016]. In my opinion, a POSA would have understood that silicone oil and liquid triglyceride both functioned as “carrier oils” that, among other things, solubilize and disperse other ingredients (id., ¶¶[0015], [0023], cl. 17, 18), and therefore would have been motivated to use even more liquid triglyceride (up to 80%) when formulating a natural deodorant stick (like in Native) that did not include, e.g., 20-40% silicone oil, to meet what the '647 Patent admits was an existing consumer preference for “more natural products.” (Ex. 1001, 1:25-26).

Thus, a POSA would have been motivated to use at least about 60% of a liquid triglyceride in a deodorant stick, and would have had a reasonable expectation of success in doing so, inasmuch as (i) Bianchi '254 already suggests a 40% amount, and (ii) a POSA would have, based on Bianchi '254's overall oil teachings, used even more of the liquid triglyceride (up to 80%) when making a stick with only that oil carrier (as taught by Native).

Ex. 1002 ¶¶ 130–131. We agree with this assessment of the combination of the teachings of Bianchi '254 and Native. A POSA would have increased the amount of liquid triglyceride in Bianchi '254 in the 40 to 80% range for the total amount of oil disclosed in Bianchi '254 when applied to the Native formulation that is natural and does not contain any silicone oil to arrive at the “at least 60% by weight liquid triglyceride” limitation in claim 16.

We also agree that this conclusion is supported by the overlapping ranges for oil disclosed in Bianchi '254 and claim 16. *See* Ex. 1002 ¶ 132.

Further, we agree with Dr. Michniak-Kohn that “there was nothing critical or unexpected about the claimed range relative to Bianchi ’254’s ranges. As a POSA would have understood, the minor deviations in the ranges yield predictable variations in degree and do not produce a different kind of result. For example, using more liquid triglyceride generally yields a softer stick, merely changing the degree of hardness.” *Id.* ¶ 132. Patent Owner offered no rebuttal to this assessment that changing the amount of liquid triglyceride is merely a variation in degree and not result. *See KSR*, 550 U.S. at 416 (stating “combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results”).

We determine that Petitioner has shown by a preponderance of the evidence that claims 16 and 17 would have been obvious over Native and Bianchi ’254.

2. Ground 6 – Native, Bianchi ’254, and Easy Homemade

Petitioner asserts that claims 1, 2, 4–7, 9–12, 18, and 19 would have been obvious over Native in view of Bianchi ’254 and Easy Homemade. Pet. 53–63. Petitioner relies on the explanation of how the limitations of claims 16 and 17 are met by the teachings of Native and Bianchi ’254, but notes that “[n]either Native nor Bianchi ’254 expressly teach a deodorant stick composition (1) substantially free of baking soda; (2) comprising an antimicrobial a power with a water solubility of at most about 50 or 90 g/L at 25°C; or (3) comprising magnesium hydroxide as the antimicrobial.” Pet. 53.

Petitioner asserts that Native’s antimicrobial, baking soda, was known to cause skin irritation for some users, and Easy Homemade explains that magnesium hydroxide with a water solubility within the claimed range is an antimicrobial that works just as well as baking soda. Pet. 53 (citing

Ex. 1009, 6, 9). Petitioner further relies on Bianchi '254 to “establish the claimed hardness range was standard.” *Id.* at 54 (citing Ex. 1002 ¶¶ 141–144).

Many of the issues raised by Patent Owner in respond to Ground 6 were also raised with regard to Ground 5, and we rely on our analysis set forth in Ground 5 above, *see supra* Section II.D.1. Patent Owner responds further that “Petitioner cannot show its Native combination discloses the hardness range required by claims 1–13 and 18–19 of the '647 Patent.” PO Resp. 54. Patent Owner explains that “Petitioner relies on the same false equivalence of Bianchi-254’s hardness measurements to those of the '647 Patent as it does with respect to Bianchi-518. *Id.* at 54 (citing Ex. 2011 at ¶¶ 119–21). We have previously addressed this issue with regard to the Lesniak grounds and rely on our analysis for that ground here. We find this argument unpersuasive to the reasons provided above in the Lesniak grounds. *See* Section II.C.2.a)(2).

Patent Owner offers the following additional criticisms of Ground 6. Patent Owner asserts that because Native does not discuss whether the deodorant is too hard or soft, “a POSA would not have known which ingredients to alter or by how much.” PO Resp. 55–56 (citing Ex. 2011 ¶ 123). Again, Petitioner does not rely on the teachings of Bianchi '254 to alter Native, but to fill in the gaps where Native is silent about the hardness of its deodorant stick. *See* Pet. 54. Petitioner relies on Bianchi '254 to “establish the claimed hardness range was standard.” *Id.* Therefore, we find Patent Owner’s argument unpersuasive.

Patent Owner also asserts that “[t]he Petition is silent on Bianchi-254’s use of aluminum salts or complexes, and neither Petitioner nor its expert explain how, or whether, a POSA could have achieved the hardness

of Bianchi-254's aluminum-based compositions, let alone the hardness claims 1-13 and 18-19, in an aluminum-free deodorant composition." PO Resp. 57 (citing Ex. 2011 ¶ 124). It is not clear from this bald assertion that aluminum affects hardness.

Relying on Dr. Michniak-Kohn's testimony, Petitioner responds that "a POSA could achieve Bianchi-254's hardness (regardless of aluminum) by simply increasing the amount of Native's structurants (e.g., stearyl alcohol) to increase hardness, or decreasing those ingredients to decrease hardness." Reply 21 (citing Ex. 1086 ¶ 6). Dr. Michniak-Kohn cites to a reference that he characterizes as "explaining that a POSA could 'easily' adjust waxes to achieve desired 'soft[ness]' based on balancing known considerations, with '[h]igher levels of wax lead[ing] to lower wear rates, and lower levels of wax tend[ing] to give higher wear rates and a softer, less desirable structure." Ex. 1086 ¶ 6 (quoting Ex. 1041, 2:64–68, 4:37–44).

We credit Dr. Michniak-Kohn's testimony as consistent with the record evidence. Petitioner has established that "[a] POSA would have known how to vary the amount of structurant (e.g., beeswax) to alter hardness (more=harder; less=softer), irrespective of the presence or absence of aluminum or non-natural ingredients." *See* Ex. 1086 ¶ 6.

Patent Owner also asserts that Petitioner has failed to show that a POSA combining the teachings of Native and Bianchi '254 would have a reasonable expectation of success. PO Resp. 57–58. Patent Owner asserts that "[a] POSA would have appreciated that Native's ingredient list suggests it is an anhydrous, wax-structured semisolid with a triglyceride-based carrier system, similar to Lesniak's examples, where even the slightest change in ingredients could yield unstable and unusable compositions." *Id.* at 58 (citing Ex. 2011 ¶ 125, Ex. 1003, Table 2).

We previously addressed Patent Owner’s assertions with respect to alleged stability issues with Lesniak’s formulations and found these arguments unpersuasive. *See supra* Section II.C.1.d). As we previously found, we find Dr. Michniak-Kohn’s testimony regarding the stability of Lesniak’s formulations persuasive. He testifies as follows.

Lesniak teaches how to make stable compositions by including a secondary wax, such as a castor wax, with a sufficiently high melting point. Ex. 1003, 1:39-54, 5:23-32, Table 2. Both of the compositions (nos. 8 and 11) tested by Lesniak that had such a wax passed the stability tests. *Id.*

Further, the stability at issue in Lesniak relates to stability at “high” temperatures, such as 49°C (120°F). *See* Ex. 1003, Table 2. Such high conditions are not a requirement to achieve a deodorant that is stable in normal conditions.

Ex. 1086 ¶¶ 7–8.

Patent Owner’s final arguments against Ground 6 involve the public accessibility of Easy Homemade and whether Easy Homemade teaches the removal and substitution of baking soda. *See* PO Resp. 58–62. Patent Owner asserts that:

Petitioner here does not provide any evidence that interested persons exercising reasonable diligence would have been able to locate Easy Homemade at the time of the priority date of the challenged patent, such as “evidence, or even argument, that [the] webpage was ‘indexed . . . (through search engines or otherwise)’ and thus locatable by a search engine.”

Petitioner’s only “evidence” of alleged public accessibility is a statement from its expert that Easy Homemade was “part of a series of related ‘Soap Deli News’ articles . . . with significant engagement from individuals globally, where she cites three different ‘Soap Deli News’ posts in apparent support. These additional blogposts do nothing to show the public accessibility of the specific Easy Homemade blogpost on which Petitioner relies, at any time before the priority date of the challenged patent.

PO Resp. 59 (citing *First Solar Inc. v. Rovshan Sade*, IPR2023-00827, Paper 13, *18 (PTAB Nov. 16, 2023); Ex. 1002 ¶ 113).

Petitioner responds that the United States Court of Appeals for the Federal Circuit (“Federal Circuit”) declined to adopt an indexing requirement for public accessibility. Reply 23 (citing *Voter Verified, Inc. v. Premier Election Soln’s*, 698 F.3d 1374, 1380 (Fed. Cir. 2012)). Petitioner asserts that in accordance with *Voter Verified*, it has shown that Easy Homemade “contains reviews from interested persons that show such persons not only could have located—but actually did locate—the website, confirming it public accessibility.” Reply 24 (citing Ex. 1009, 7–8; Ex. 1016, 7–8, Ex. 1018, 6).

We determine that Petitioner has shown by a preponderance of the evidence that Easy Homemade was publicly accessible.

A reference is considered publicly accessible if it is “made available to the extent that persons interested and ordinarily skilled in the subject matter or art, exercising reasonable diligence, can locate it.” *Jazz Pharms., Inc. v. Amneal Pharms., LLC*, 895 F.3d 1347, 1355 (Fed. Cir. 2018). To “determine whether an interested researcher would have been sufficiently capable of finding the reference and examining its contents,” we “must consider all of the facts and circumstances surrounding the disclosure.” *In re Lister*, 583 F.3d 1307, 1312 (Fed. Cir. 2009); *see also In re Klopfenstein*, 380 F.3d 1345, 1350 (Fed. Cir. 2004) (explaining that a determination of whether a reference qualifies as a printed publication “involves a case-by-case inquiry into the facts and circumstances surrounding the reference’s disclosure to members of the public”).

Patent Owner’s main argument appears to be that Soap Daily News was not indexed. *See* PO Resp. 59. As Petitioner points out, however, such

evidence is not necessarily required in all cases. *See Sandoz Inc. v. Abbvie Biotechnology LTD.*, IPR2018-00156, Paper 11, *11 (PTAB June 5, 2018) (citing *Voter Verified*, 698 F.3d at 1381). “Rather, absent evidence of indexing, testimony indicating that the particular online publication or website on which the reference was published was well-known to the community interested in subject matter of the reference, and that, upon accessing the website, those interested would have found the reference using the website’s own search functions can support the ultimate determination that a given reference was publicly accessible.” *Id.* at 11–12 (citing *Voter Verified*, 698 F.3d at 1380–81).

Petitioner presents evidence from the Wayback Machine for the Easy Homemade Blogpost, *see* Ex. 1009, 1–2, in addition to three other Soap Deli News posts, *see* Exs. 1017–1019. Dr. Mishniak-Kohn testifies “Easy Homemade was widely accessible, being part of a series of related “Soap Deli News” articles dedicated to deodorant formulations with significant engagement from individuals globally.” Ex. 1002 ¶ 143 (citing Ex. 1009; Exs. 1016–1018). Dr. Mishniak-Kohn also testifies that “a POSA would have appreciated that such websites from independent formulators striving for more natural options present an indication of consumer preferences, which a POSA would desire to meet in formulating new commercial products.” *Id.* We credit Dr. Mishniak-Kohn’s testimony here, and determine that Easy Homemade was publicly accessible and qualifies as a printed publication.

Patent Owner also asserts that Petitioner misreads Easy Homemade as teaching the removal and substitution of baking soda, but “[r]ather, the Easy Homemade hobbyist tells her followers to purchase a pre-made baking soda-free deodorant base—a base Petitioner’s expert could not identify the

ingredients of—and add some magnesium hydroxide powder to it. As Petitioner’s expert confirms, Easy Homemade ‘doesn’t talk about removing any ingredients from these sticks; she’s talking about adding ingredients.’” PO Resp. 60 (citing Ex. 1009, 7, Ex. 1012, 120:6-8, 121:1–9; Ex. 2011 ¶¶ 127–28).

Petitioner responds that using prior art elements according to their established functions is obvious. Reply 22–23 (citing *KSR*, 550 U.S. at 417; *Intel Corp. v. PACT XPP Schweiz AG*, 61 F.4th 1373, 1380-81 (Fed. Cir. 2023)). Therefore, Petitioner asserts that “[a] known antimicrobial for deodorants, baking soda, caused irritation for some users, and another known antimicrobial for deodorants, magnesium hydroxide, ameliorated that issue when substituted for baking soda. ‘Nothing more is required to show a motivation to combine under *KSR*.’” *Id.* at 23 (citing *Intel*, 61 F.4th at 1381). We agree.

Dr. Michniak-Kohn points to evidence that “a POSA would have known that deodorants containing baking soda could cause skin irritation for some users and Easy Homemade confirms what POSA would have known—namely, that magnesium hydroxide was a known substitute for baking soda that avoided skin irritation.” Ex. 1002 ¶ 145 (citing Ex. 1009, 6, 9).

Petitioner further notes that Easy Homemade expressly discusses substituting magnesium hydroxide for baking soda because Easy Homemade states that “baking soda can easily be swapped for magnesium hydroxide.” Reply 23 (citing Ex. 1009, 6, 9). We agree with Petitioner that “Easy Homemade just directs the POSA to substitute one known antimicrobial for another.” *Id.*

Based on the full record before us, we determine that Petitioner has shown by a preponderance of the evidence that claims 1, 2, 4–7, 9–12, 18,

and 19 would be been obvious over Native, Bianchi '254, and Easy Homemade.

3. *Ground 7 – Native, Bianchi '254, Easy Homemade, and Nabavi*

Petitioner asserts that claims 3 and 13 would have been obvious over Native, Bianchi '254, Easy Homemade, and Nabavi. Pet. 63–66. Petitioner relies on Nabavi to teach a deodorant stick with cinnamon essential oil that Petitioner asserts satisfies the additional limitations set forth in claims 3 and 13. *Id.*

Patent Owner responds that a POSA would not have looked to Nabavi because “Nabavi does not disclose cinnamon essential oil as an underarm deodorant antimicrobial, and a POSA would not have been motivated to use it as such, particularly given Nabavi’s teachings of cinnamon as a skin irritant.” PO Resp. 62 (citing Ex. 1067, 7742; Ex. 2011 ¶ 133–34). Patent Owner also asserts that “[i]f, according to Petitioner, the POSA would have replaced baking soda with magnesium hydroxide in the Native combination, then Petitioner’s motivations to further combine with Nabavi—that rest on the POSA being motivated to, again, replace backing soda in Native (*see* Petition at 64)—would not exist.” *Id.* at 63 (citing Ex. 2011 ¶ 135).

Dr. Michniak-Kohn testifies:

In my opinion, the combination of Native in view of Bianchi '254, Easy Homemade, and Nabavi renders claim 3 obvious. In particular, Native teaches deodorant stick compositions containing baking soda (a known antimicrobial) and essential oils, but it does not expressly teach deodorant stick compositions containing magnesium hydroxide or cinnamon oil as an antimicrobial. As discussed above, it would have been obvious to substitute magnesium hydroxide for baking soda as Native’s primary antimicrobial. It also would have been obvious to include an additional antimicrobial such as cinnamon essential oil taught by Nabavi, to combat bacteria

thought to cause odor and product degradation, as well as to provide a pleasant fragrance. Ex. 1067, 7733, 7739, 7742. A POSA would have used cinnamon essential oil in a lower amount than the primary antimicrobial because it was thought to cause skin irritation. *Id.*, 7742.

Ex. 1002 ¶ 187.

We credit Dr. Michniak-Kohn's testimony here as consistent with the record evidence. Adding cinnamon essential oil to combat bacteria in a lower amount to prevent skin irritation would be using such oil as described by Nabavi as having both aromatic and antibacterial effects. *See* Ex. 1002 ¶¶ 43, 110, 112 (stating cinnamon use in "cosmetics could be a good strategy to reduce or avoid bacterial degradation"), 113 (describes using cinnamon essential oil because it would serve the dual purpose of being an antimicrobial and a fragrance), 183. We find that addition of cinnamon essential oil to the claimed formulation to be merely a combination of familiar elements according to known methods that yield predictable results, and therefore, would have been obvious. *See KSR*, 550 U.S. at 416.

Based on the full record before us, we determine that Petitioner has shown by a preponderance of the evidence that claims 3 and 13 would be been obvious over Native, Bianchi '254, Easy Homemade, and Nabavi.

III. CONCLUSION⁷

Based on the fully developed trial record, Petitioner has demonstrated by a preponderance of the evidence that claims 1–7, 9–13, and 16–19 of the '647 patent are unpatentable.

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/ Basis	Claim(s) Shown Unpatentable	Claim(s) Not Shown Unpatentable
16, 17	102/103	Lesniak	16, 17	
1, 4–7, 9–12, 18–19	103	Lesniak, Bianchi '518	1, 4–7, 9–12, 18–19	
2	103	Lesniak, Bianchi '518, Phinney	2	
3, 13	103	Lesniak, Bianchi '518, Phinney, Nabavi	3, 13	
16, 17	103	Native, Bianchi '254	16, 17	
1–2, 4–7, 9–12, 18–19	103	Native, Bianchi '254, Easy Homemade	1–2, 4–7, 9–12, 18–19	
3, 13	103	Native, Bianchi '254, Easy Homemade, Nabavi	3, 13	
Overall Outcome			1–7, 9–13, 16–19	

⁷ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

IV. ORDER

In consideration of the foregoing, it is

ORDERED that claims 1–7, 9–13, 16–19 of the '647 patent have been shown to be unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

FOR PETITIONER:

Justin Oliver
Sarah Brooks
VENABLE LLP
joliver@venable.com
ssbrooks@venable.com

FOR PATENT OWNER:

David Maiorana
Marlee Hartenstein
Jesse Wynn
JONES DAY
dmaiorana@jonesday.com
mhartenstein@jonesday.com
jwynn@jonesday.com